

The DR–DCR Arc, 2026-07-26 → 2026-07-27: A Reader's Guide

Human director: Jawaun Brown **Producing agent:** Claude Code, directed **Program:** Deletion-Repair – synthesis **Date:** 2026-07-27

Abstract

Between 2026-07-26 and 2026-07-27, twelve papers were shipped across two research arcs – deletion-repair (DR1–DR5b), date-cut retrodiction (DCR1–DCR2a), and the code-correctness corollary triple (DR6/DR6c/DR6d). Every paper preregistered its gates, honored its verdict, and left its predecessors byte-identical. The arcs converged on a single structural result: **DR5, a theorem about the verification of multi-realisation commitments**, and its empirical triangulation across two subject-matter domains and two verifier architectures.

This synthesis exists so that anyone reading the arc – including future me – can pick up the story cold. It is not a new experiment. It maps the terrain the twelve papers laid down and points at the natural follow-ups.

1. The problem the arc set out to answer

Can an execution-free "nominator" – a scoring function that ranks propositions in a scientific corpus – identify the specific commitment a community should delete? In the DCR arc's concrete instance: given only what was public before 1905, can a nominator rank the commitment Einstein actually deleted (roughly, absolute simultaneity) above alternatives?

The question is important because deletion-repair is the framework the DR arc has been building for years; the DCR corpus was the first real empirical setting, with ground truth (Einstein's 1905 paper) attached.

The answer, developed across the arc, is: **the question was underspecified**. Einstein did not delete a proposition. He deleted a *class* of propositions, realised in the corpus under six or more non-equivalent surface forms. That distinction, invisible when the target commitment is treated as a single retrievable object, is the structural finding the arc landed at.

2. What each paper contributed

DR1–DR4 (before this session): the toy-problem foundation. Nominator design, cost functions, disjunctive combinators, calibrated base rates. DR4 (this session's opening paper) closed the toy arc with all four gates G0.

DCR1 (PR #422): built the pre-1905 electrodynamics corpus. Fifteen public-domain documents from Maxwell 1865 to Lorentz 1904, ~496k characters, checksummed, chrome-stripped structurally. Introduced relational vocabulary residue (a token is residue iff the extractor emits it AND the corpus lacks it at that cut) – the DCR arc's key methodological artifact.

DCR1b (PR #423): read every hit. Adjudicated the DCR1 verdict at the individual-proposition level; found T2 (privileged frame) matched via three failure modes (polarity, referent, label) requiring vetoes. Introduced the "read every hit" discipline that later prevented DCR1's exemplar-only adjudication from over-stating by 27%.

DCR1c (PR #424): all six gates G0 on stated-commitment extraction. The 1880 deep placebo matched zero propositions to any target facet under three independent extraction passes – Spencer's candidate-selection circularity tested directly and absent. But T1 (absolute simultaneity) matched zero at every cut. The paper refused to decide between "matcher too strict" and "commitment absent from corpus," and licensed DCR1d as the discriminating experiment.

DCR1d (PR #425): positive control on Newton's Scholium. T1 matcher fires three times on Newton's explicit "absolute time." Reading closed: DCR1c's T1 absence is *historical*, not instrumental. Presuppositions have a life cycle: explicit → assumed → invisible. Einstein's deletion was available exactly because absolute simultaneity had gone silent by 1900.

DCR1e (PR #426): a presupposition-inferring extractor over the same corpus. Surfaced T1 content in five documents (Larmor 1900's "common time t in which...", Lodge 1897's "definite and independent of the motion", Maxwell 1865's "instant across the whole medium", Poincaré 1898 twice). **target_v3 rejected all five.** The recognizer gap discovered.

DCR1f (PR #427): a matcher successor (target_v4) widened for presuppositional phrasings. Fires on 7 T1 propositions at 1904, but **also on Maxwell at 1880**, and fails held-out validation at 32.5%. Instrument reached its precision-recall wall. §4 identified T1 as a *class* rather than a point, across six registers.

DR5 (PR #428, joint with DCR2a): the theorem. Proposition-ranking nominator N cannot distinguish D from any r_i when $|\text{realisations}(D)| > 1$; escape requires importing an external grouping function g; §5 states the wider claim about verification.

DCR2a (PR #428): empirical companion. Class-based scoring over the DCR1e consensus. T1 class ranks at position 3 at 1904 (proposition-blind best T1 was position 4) – a real one-position gain. **But T1 class also ranks 2 at the 1880 placebo** because a single misfiring member inflates low-cardinality classes. Class-aware nomination is a strict generalisation but does not delete the ambiguity – it moves it from N to g's soundness, as DR5 predicts.

DR5b (PR #429): verification corollaries. Enumerates four non-linguistic domains where DR5 operates (code correctness, LLM reasoning verification, retrieval-augmented grounding, latent goal identification). §7 gives six diagnostic questions any verifier designer can use to check whether their protocol is above or below the wall.

DR6 (PR #430): first empirical DR5 test in code correctness. Naive-UTC datetime commitment, 5 realisations + 5 placebos, 3 sandboxed Claude verifiers. **Wall did not appear.** Realisations 8–10, placebos 0–1, overlap gap +7. Sharpened DR5: wall bites when (a) D has no canonical form OR (b) verifier can't reason semantically.

DR6c (PR #431): same D, regex verifier with one pattern per enumerated realisation surface form. Perfect discrimination (all realisations 2, all placebos 0), wall did not appear. **Re-refined DR5:** condition (b) is not just semantic reasoning; it is any means of "accessing the canonical form" including complete enumeration.

DR6d (PR #432): same regex, one novel realisation added at test time. **R6 = 0. Wall bit.** Overlap gap 0 = 0. Triangulation complete.

3. The theorem, in one paragraph

Let D be a commitment with realisations $\{r_1, \dots, r_k\}$ in a corpus C, $k > 1$. Let $N: C \rightarrow \mathbb{R}$ be a proposition-ranking nominator ($N(p)$ depends only on p). Then N assigns k different scores to the realisations; no aggregation into a single "D-score" is derivable from N alone; escape requires an external grouping function g that identifies realisations(D, C). Once g is imported, ranking is over classes, not propositions.

Refined empirical condition (DR6d): the wall bites when both
enumerated by the verifier designer, and
reasoning about D itself.

- (a) the realisation set is open-ended at test time – some r_i was not
- (b) the verifier cannot reach unseen realisations via semantic

Both must fail. If either holds – either the verifier enumerates the realisation set fully in advance (DR6c) OR reasons semantically about D (DR6, DR6e-predicted) – the wall does not appear.

4. Where the wall bit and where it did not

paper	corpus	verifier	(a) open?	(b) semantic?	wall?
DCR1c	pre-1905 physics	stated-commitment extractor + target_v3	closed	no	didn't ask (T1 absent)
DCR1e	pre-1905 physics	presupposition extractor + target_v3	open	no	bit (recognizer gap)
DCR1f	pre-1905 physics	presup extractor + target_v4	open	no	bit (placebo fires)
DR6	2026 Python	Claude subagents	closed	yes	did not bite
DR6c	2026 Python	regex, 5 patterns/5 realisations	closed by construction	no	did not bite
DR6d	2026 Python	same regex + R6 novel	open	no	bit

DR5 predicts every row. Every row is what actually happened.

5. What the arc has and has not established

Established:

provenance guarantees (relational residue, structural chrome removal, translation-risk exoneration).

discipline (three passes, 2-of-3 consensus, no-file-access sandbox, read-every-hit adjudication).

matter domains and two verifier architectures.

DR6–DR6d, each honored. Nothing tuned after seeing results.

- A methodology for building a real domain-specific corpus with
- A methodology for sandboxed subagent extraction with sound
- A theorem (DR5) with an empirical triangulation across two subject-
- Six preregistered gates across DCR1c–DCR2a, each honored; six across

Not established:

real material. DCR2a tested only class-vs-proposition scoring, not the DR1–DR4 nominator pipeline.

would rank the T1 class first. DCR2a found target_v4 is not sound at the 1880 cut; constructing a sound g is an open problem.

correctness and pre-1905 physics.

- Whether the DR-arc's deletion-repair *nominator* actually works on
- Whether class-aware nomination on the DCR corpus, with a *sound g*,
- Whether the DR5 wall applies to open-corpus domains beyond code

6. Natural follow-ups (all named in one paper or another)

Extension of the empirical triangulation:

catches R6 semantically, wall stays down.

reappears because the verifier cannot access the canonical form.

- **DR6e** – Claude verifier on DR6d's 11 snippets. Predicted: Claude

- **DR6f** – Claude verifier without D in the prompt. Predicted: wall

Theorem-shape work:

must g have to be sound and complete for a specific D? Under what conditions can g be learned without collapsing back to a proposition-ranking problem (Spencer's circularity one level up)?

- **DR7** – grouping function correctness. What structural properties

Companion empirical work:

DCR2a's cardinality-based rules failed the placebo; a rule that requires multi-document support may not.

than as an intermediate step feeding target_v4.

- **DCR2b** – multi-document coverage aggregation rule for class scoring.
- **DCR2c** – presupposition-inferring extractor as g itself, rather

None of these is required to establish the DR5 result. Each would sharpen a specific boundary further.

7. Discipline summary

For any reader building their own version of this arc:

after run); the paper reported the slip. Every subsequent run honored the discipline. DR6 held it strictly.

Prevents post-hoc threshold-fitting.

edits prior modules. All twelve prior runners produce byte-identical numbers on re-run.

The DCR1e pass-1 breach (one agent read the repo's own code) was detected and rerun; comparison between the breached and clean passes quantified the effect (67.7% content agreement, no facet-level change).

are drafted. DCR1f's target_v4 was tuned to DCR1e outputs; DCR1f's R2 32.5% failure on held-out showed this was insufficient. DR6d avoided the same trap by drafting R6 from a label before the code was written.

- Preregister before every run. DCR1c slipped once (prereg written
- Every threshold as imported constant, not restated in the runner.
- Every new experiment additive: a new module, a new runner, never
- Sandboxed subagents forbid file access beyond the named document.
- Held-out validation sets committed at fixed SHA-256 BEFORE matchers

Ten discipline notes, one theorem, twelve papers. The arc is intended to be reproducible in full from main at any commit hash from the 2026-07-26 → 2026-07-27 window.

Appendix: reading order

For someone approaching cold and wanting the shortest path to the finding:

1. **DR5** – the theorem, everything else motivates or corroborates it.
2. **DCR1f** – the empirical wall on a real corpus, with ground truth.
3. **DR6 + DR6c + DR6d** – the triangulation on code.
4. **DCR2a** – what class-aware nomination inherits.
5. **DR5b** – practical guidance for anyone building verifiers.

For depth on the DCR corpus specifically:

1. **DCR1** – corpus construction and relational residue.
2. **DCR1c** – six-gate G0, T1 absent.
3. **DCR1d** – Newton positive control (T1 absence is historical).
4. **DCR1e** – presupposition extractor, recognizer gap.
5. **DCR1f** – target_v4, placebo fires.

All papers reproduce via each runner's own `run_dcr1x.py` (`x` in `{"", b, c, d, e, f}`) or `run_dr6x.py` (`x` in `{"", c, d}`). Prior verdicts stay byte-identical after every subsequent paper.